

VIBRATION OF STRUCTURES (ME60428) ASSIGNMENT – 3 BY SOUVIK SARKAR (22ME10083)

Derivation of the BVP and Solving the particular solution with detailed explanation with Eigen Function and Green function scheme →

(row 01)
Date: 19/02/2024

VoS - Assignment-3 by 22ME10083

1. Sliding - fixed steel string, $d = 5\text{mm}$

a) $q(x,t) = q_0 \cos \Omega t$, k^{th} eigen value and functions are -

$\omega_k = \frac{(2k-1)\pi c}{2l}$, $W_k(x) = \cos\left(\frac{(2k-1)\pi x}{2l}\right)$

$l = 2\text{m}$, $\rho = 7850\text{ kg/m}^3$, $T = 1000\text{ N}$,
 $q_0 = 60\text{ N}$, $\Omega = 220\text{ rad/s}$.

for string $\rho(x) = \rho A(x)$; $K[W] = -[TW_x]_{x=l}$ so, the Equation of Motion becomes

$\rho A(x) W_{,tt} - [TW_x]_{x=l} = -q_0 \cos \Omega t \Rightarrow [W_{,tt} - c^2 W_{,xx} = -q_0 \cos \Omega t]$ (i)

where, $c = \sqrt{\frac{T}{\rho A}}$ & $q_0 = \frac{Q_0}{\rho A l}$

Now, assume this Equation (i) has a solution which is comprising the complementary (homogeneous) & particular Integral solution. Assume, $W(x,t) = w_H(x,t) + w_P(x,t)$ (ii)

Now, we know $w_H(x,t)$ satisfy the free vibration or Modal Analysis solution. where, $[w_H]_{,tt} - c^2 w_H]_{,xx} = 0$ (iii)

Also, the particular solution will be scaling factor of the driving frequency. term assume $[w_P = X(x) \cos \Omega t]$ (iv)

Now, putting Eqn (ii) in the Differential Eqn (i) we get,

$w_H]_{,tt} + w_P]_{,tt} - c^2 w_H]_{,xx} - c^2 w_P]_{,xx} = -q_0 \cos \Omega t$

from Eqn (iii) we get $[w_P]_{,tt} - c^2 w_P]_{,xx} = -q_0 \cos \Omega t$ (v)

put Eqn (iv) into Differential Eqn (v) we get,

$-\Omega^2 X(x) \cos \Omega t - c^2 X''(x) \cos \Omega t = q_0 \cos \Omega t$

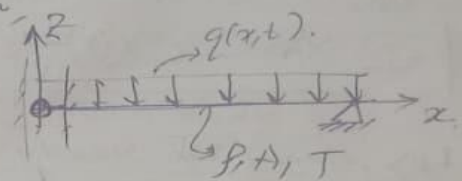
for getting non-trivial solution $\cos \Omega t \neq 0$ so, we get,

$-\Omega^2 X(x) - c^2 X''(x) = q_0$ (vi)

Now, at $x=0$, $TW_x|_{x=0} = 0 \Rightarrow \frac{\partial}{\partial x}(w_H + w_P)|_{x=0} = 0$

as w_H & w_P are individually solutions of the differential Equation so, $\frac{\partial w_H}{\partial x}|_{x=0} = 0$ & $\frac{\partial w_P}{\partial x}|_{x=0} = 0 \Rightarrow X'(0) \cos \Omega t = 0$

so, @ $x=0$, $X'(0) = 0 \rightarrow$ Sliding side BCs (Dynamic) BC



@ $x=l$, $w(l,t)=0 \Rightarrow w_p(l,t) = X(l)\cos \Omega t = 0$
 And, $\cos \Omega t \neq 0$ (Non trivial solution so, $X(l)=0$)

$[-\Omega^2 X(x) - c^2 X''(x) = q_0 \text{ \& } X(0)=0 \text{ \& } X(l)=0]$ — (v)

$[-\Omega^2 X(x) + c^2 X''(x) = q_0]$ — $\rightarrow$ Boundary value Problem

1.b) Analytical Steady State Response:-

Let's assume homogeneous solution $X_H(x)$ & particular solution is $X_P(x)$

Now, $X''(x) + \left(\frac{\Omega}{c}\right)^2 X(x) = +\frac{q_0}{c^2}$ — $\rightarrow$ Steady state differential eqn

$\therefore X_H(x) = M \cos\left(\frac{\Omega}{c}x\right) + N \sin\left(\frac{\Omega}{c}x\right)$, and the particular solution be $X_P(x) = G$ Now $X_P(x) = \frac{+q_0}{c^2 - (\Omega^2/c^2)}$
 $\left(\frac{\Omega}{c}\right)^2 G = +\frac{q_0}{c^2} \Rightarrow G = \frac{+q_0}{\Omega^2}$

$\therefore$ Steady state solution $X(x) = M \cos\left(\frac{\Omega}{c}x\right) + N \sin\left(\frac{\Omega}{c}x\right) + \frac{q_0}{\Omega^2}$

Applying BCs

(i) $X'(0) = N \cdot \frac{\Omega}{c} \cdot 1 = 0 \Rightarrow N = 0$

(ii) $X(l) = 0 \Rightarrow X(l) = M \cos\left(\frac{\Omega}{c}l\right) + \frac{q_0}{\Omega^2} = 0$

$\therefore M = \frac{-q_0}{\Omega^2 \cos\left(\frac{\Omega}{c}l\right)}$

$X(x) = \frac{-q_0}{\Omega^2} \left[\frac{\cos\left(\frac{\Omega}{c}x\right)}{\cos\left(\frac{\Omega}{c}l\right)} - 1 \right]$ — (vi)

So, steady state solution is $w_p(x,t) = \frac{-q_0}{\rho A \Omega^2} \left[\frac{\cos\left(\frac{\Omega}{c}x\right)}{\cos\left(\frac{\Omega}{c}l\right)} - 1 \right] \cos(\Omega t)$
 as $q_0 = \frac{Q_0}{\rho A L}$

1.c) Eigen function Expansion Method:- assume $X(x) = \sum_{k=1}^{\infty} \alpha_k W_k$
 where α_k are scalar multiples, $W_k \rightarrow$ mode shapes $\rightarrow$ obtained from EVP
 from eqn (v) put $X(x)$ value we get,

$-\Omega^2 \sum_{k=1}^{\infty} \alpha_k W_k - c^2 \sum_{k=1}^{\infty} \alpha_k W_k'' = -q_0$

from the eigen value problem we get, $-\omega_k^2 W_k - c^2 W_k'' = 0$
 so our eqn become, $-\sum_{k=1}^{\infty} \Omega^2 \alpha_k W_k + \sum_{k=1}^{\infty} \alpha_k \omega_k^2 W_k = -q_0$

so, $\alpha_k \left[\sum_{k=1}^{\infty} (\omega_k^2 - \Omega^2) \alpha_k W_k \right] = -q_0$ — (vii)

Taking inner product with W_j where every term except $j=k$ will cancel out so, we get $\alpha_j = \frac{\int_0^L Q_0 W_j(x) dx}{\int_0^L W_j^2 dx (\omega_j^2 - \Omega^2)}$
 where $j=1, 2, 3, \dots$

Now $W_j(x) = \cos\left(\frac{(2j-1)\pi x}{2L}\right)$
 $\omega_j = (2j-1)\frac{\pi c}{2L} \Rightarrow \alpha_j = \frac{-Q_0 \int_0^L \cos\left(\frac{(2j-1)\pi x}{2L}\right) dx}{(\omega_j^2 - \Omega^2) \int_0^L \cos^2\left(\frac{(2j-1)\pi x}{2L}\right) dx}$

$$\alpha_j = \frac{-Q_0 \cdot 2L \cos(j\pi)}{(\pi - 2j\pi)(\omega_j^2 - \Omega^2) \cdot L/2}$$

$$\therefore \alpha_j = \frac{4Q_0 \cos(j\pi)}{\pi L (2j-1)\pi (\omega_j^2 - \Omega^2)}$$

where $\omega_j = (2j-1)\frac{\pi c}{2L}$ and $Q_0 \rightarrow$ force Amplitude per unit length.

So, Steady State Eppr. $w_p^E(x, t) = \left(\sum_{k=1}^{\infty} \alpha_k W_k(x) \right) \cos \Omega t$
 $w_p^E(x, t) = \left(\sum_{k=1}^{\infty} \frac{4Q_0 \cos\left(\frac{(2k-1)\pi x}{2L}\right)}{\pi L (2k-1)\pi \left[\frac{(2k-1)^2 \pi^2 c^2}{4L^2} - \Omega^2\right]} \right) \cos \Omega t$

1.d) Green function Approach: $-\Omega^2 X(x) - c^2 X''(x) = -Q_0$
 Let's assume $-\Omega^2 X(x) - c^2 X''(x) = \delta(x - \bar{x})$

ans. $X(x) = G(x, \bar{x}, \Omega)$ & $G'(0, \bar{x}, \Omega) = 0$ and $G(L, \bar{x}, \Omega) = 0$
 Green function of the system

B.V.P: $-\Omega^2 G(x, \bar{x}, \Omega) - c^2 G''(x, \bar{x}, \Omega) = \delta(x - \bar{x})$ (viii)
 apart from $\bar{x}$ every where the load is zero.

$$G(x, \bar{x}, \Omega) = A_L \sin\left(\frac{\Omega}{c} x\right) + B_L \cos\left(\frac{\Omega}{c} x\right) \quad 0 \leq x < \bar{x}$$

$$G(x, \bar{x}, \Omega) = A_R \sin\left(\frac{\Omega}{c} x\right) + B_R \cos\left(\frac{\Omega}{c} x\right) \quad \bar{x} < x \leq L$$

Continuity Condition: $G(x \rightarrow \bar{x}^-, \bar{x}, \Omega) = G(x \rightarrow \bar{x}^+, \bar{x}, \Omega)$

Jump Condition: $\lim_{\epsilon \rightarrow 0} \int_{\bar{x}-\epsilon}^{\bar{x}+\epsilon} (-\Omega^2 G - c^2 G'') dx = \int_{\bar{x}-\epsilon}^{\bar{x}+\epsilon} \delta(x - \bar{x}) dx$

G is smooth so, over that domain all results in zero
 $\Rightarrow c^2 G_{,x} \big|_{\bar{x}-\epsilon}^{\bar{x}+\epsilon} = -1 \Rightarrow G_{,x}(\bar{x}^+, \bar{x}, \Omega) - G_{,x}(\bar{x}^-, \bar{x}, \Omega) = -\frac{1}{c^2}$

By solving in Mathematica I got

$$G(x, \bar{x}, \Omega) = \frac{1}{\Omega c} \cos\left(\frac{\Omega}{c} l\right) \begin{cases} \cos\left(\frac{\Omega}{c} x\right) \sin\left(\frac{\Omega}{c} (l - \bar{x})\right) & 0 \leq x < \bar{x} \\ \cos\left(\frac{\Omega}{c} \bar{x}\right) \sin\left(\frac{\Omega}{c} (l - x)\right) & \bar{x} < x \leq L \end{cases}$$

So, using the Green function method, the general solution is given by

$$X(x) = \int_0^L \frac{-Q_0}{PAL} G(x, \bar{x}, \Omega) d\bar{x}$$

$$X(x) = \int_0^x \frac{-Q_0}{PAL} G(\bar{x}^+, \bar{x}, \Omega) d\bar{x} + \int_x^L \frac{-Q_0}{PAL} G(\bar{x}, \bar{x}, \Omega) d\bar{x}$$

$$X(x) = \frac{-Q_0}{PAL} \left[\frac{(\cos(\frac{(L-x)\Omega}{c}) - 1) \cos(\frac{x\Omega}{c})}{\Omega^2 \cos(\frac{L\Omega}{c})} + \frac{\sin(\frac{(L-x)\Omega}{c}) \sin(\frac{x\Omega}{c})}{\Omega^2 \cos(\frac{L\Omega}{c})} \right]$$

$$\therefore u_p(x, t) = X(x) \cos \Omega t$$

$$W_p(x, t) = \frac{-Q_0 \cos \Omega t}{PAL} \left[\frac{(\cos(\frac{(L-x)\Omega}{c}) - 1) \cos(\frac{x\Omega}{c})}{\Omega^2 \cos(\frac{L\Omega}{c})} + \frac{\sin(\frac{(L-x)\Omega}{c}) \sin(\frac{x\Omega}{c})}{\Omega^2 \cos(\frac{L\Omega}{c})} \right]$$

Using Mathematica, the complex integral has been solved and code is also attached along with the report

* All the Comparison plots are plotted using Mathematica at $t=1s$, $t=50s$, $t=70seconds$ for the three cases. (Code is also attached).

Code block For finding out the eigen function expansion: using symbolic mathematic programming language

```
W[k_] := Cos[(2*k-1) * Pi * x / (2*L)];
\Omega[k_] := (2*k-1) * Pi * c / (2*L);
```

(* Modal coefficient \[Alpha] *)

```
\[Alpha][k_] := Integrate[(-Q0/(\[Rho] * a*I)) * W[k], {x, 0, l}] /
((-\[CapitalOmega]^2 + \[Omega][k]^2) * Integrate[W[k]^2, {x, 0, l}]);
\[Alpha][k]
```

```
(* Summing the expansion *)
xpm� = Sum[\[Alpha][k] * W[k], {k, 1, num}];
```

```
(* General expression used for comparison *)
wpE[x_, t_] := xpm� * Cos[\[CapitalOmega] * t]
```

Output:

Eigenfunction expansion method

$$Out[4] = \frac{247.502 \cos[k\pi]}{(1. - 2. k) (-48400 + 3996.94 (-1 + 2 k)^2) \left(1 + \frac{\sin[2k\pi]}{\pi - 2k\pi}\right)}$$

$$Out[5] = \cos[220 t] \left(0.00557399 \cos\left[\frac{\pi x}{4}\right] - 0.00663853 \cos\left[\frac{3\pi x}{4}\right] - 0.000960736 \cos\left[\frac{5\pi x}{4}\right] + 0.000239793 \cos\left[\frac{7\pi x}{4}\right] - 0.0000998731 \cos\left[\frac{9\pi x}{4}\right] + 0.0000516973 \cos\left[\frac{11\pi x}{4}\right] - 0.0000303606 \cos\left[\frac{13\pi x}{4}\right] + 0.0000193912 \cos\left[\frac{15\pi x}{4}\right] - 0.0000131551 \cos\left[\frac{17\pi x}{4}\right] + 9.34133 \times 10^{-6} \cos\left[\frac{19\pi x}{4}\right] \right)$$

Code block For finding out the Green function expansion: using symbolic mathematic programming language

(* 1. Define the Piecewise Green's Function forms *)

```
GL[x_, z_, \[CapitalOmega]] := AL * Sin[\[CapitalOmega] * x / c] + BL * Cos[\[CapitalOmega] * x / c];
```

```
GR[x_, z_, \[CapitalOmega]] := AR * Sin[\[CapitalOmega] * x / c] + BR * Cos[\[CapitalOmega] * x / c];
```

(* 2. Define the Boundary and Continuity/Jump Conditions *)

```
eq1 = (D[GL[x, z, \[CapitalOmega]], x] /. x -> 0) == 0;
```

```
eq2 = GR[l, z, \[CapitalOmega]] == 0;
```

```
eq3 = GL[z, z, \[CapitalOmega]] == GR[z, z, \[CapitalOmega]];
```

```
eq4 = (D[GR[x, z, \[CapitalOmega]], x] - D[GL[x, z, \[CapitalOmega]], x] /. x -> z) == -1/c^2;
```

(* 3. Solve for the coefficients *)

```

sol = Simplify[Solve[{eq1, eq2, eq3, eq4}, {AL, BL, AR, BR}]];

(* 4. Display the results in Symbolic/Traditional Form *)
Print["The Green's Function for 0 <= x <= z is:"];
GL[x, z, \[CapitalOmega]] /. sol[[1]] // Simplify // TraditionalForm

Print["The Green's Function for z <= x <= l is:"];
GR[x, z, \[CapitalOmega]] /. sol[[1]] // Simplify // TraditionalForm

(* Extract and Display the simplified expressions *)
finalGL = Simplify[GL[x, z, \[CapitalOmega]] /. sol[[1]]];
finalGR = Simplify[GR[x, z, \[CapitalOmega]] /. sol[[1]]];
Print["Simplified GL: ", finalGL];
Print["Simplified GR: ", finalGR];
(* Solve for Displacement using Green's Function Integration *)
xpg = (-Q0 / (\[Rho] * a*l)) * (Integrate[finalGR, {z, 0, x}] +
    Integrate[finalGL, {z, x, l}]);
wpG[x_,t_]=xpg*Cos[\[CapitalOmega]*t];
wpG[x,t]//Simplify

```

Output:

The Green's Function for $0 \leq x \leq z$ is:

Out[]//TraditionalForm=

$$\frac{\sec\left(\frac{l\Omega}{c}\right) \cos\left(\frac{x\Omega}{c}\right) \sin\left(\frac{\Omega(l-z)}{c}\right)}{c\Omega}$$

The Green's Function for $z \leq x \leq l$ is:

Out[]//TraditionalForm=

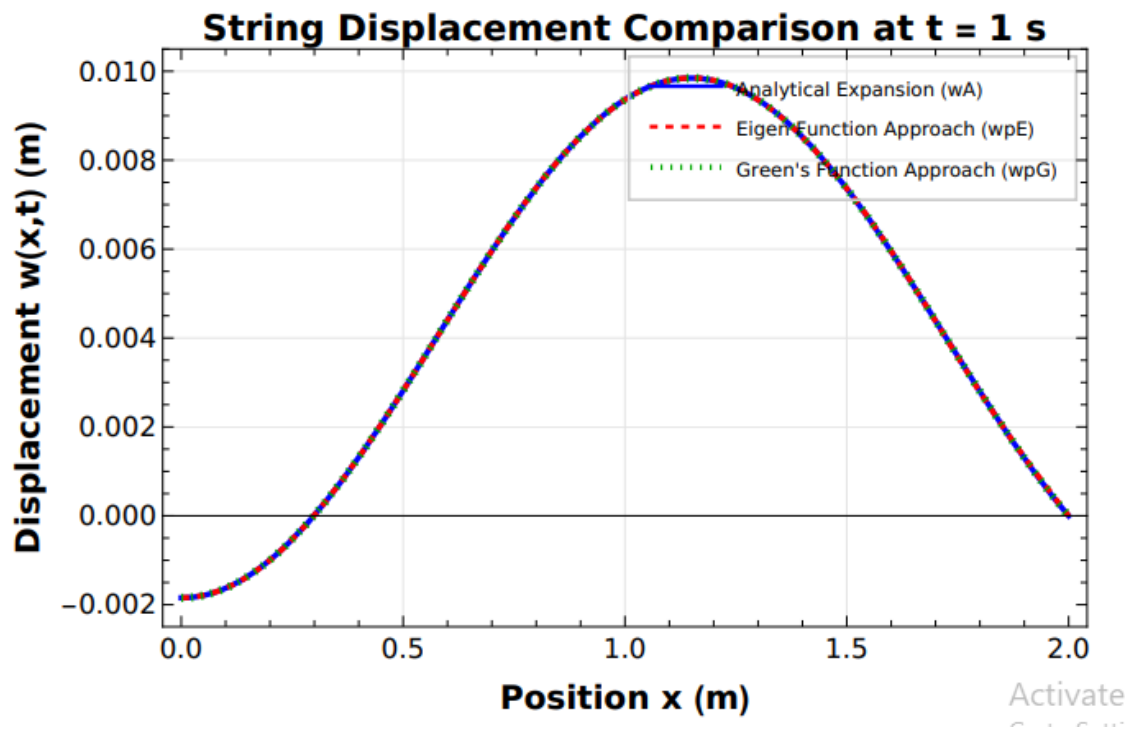
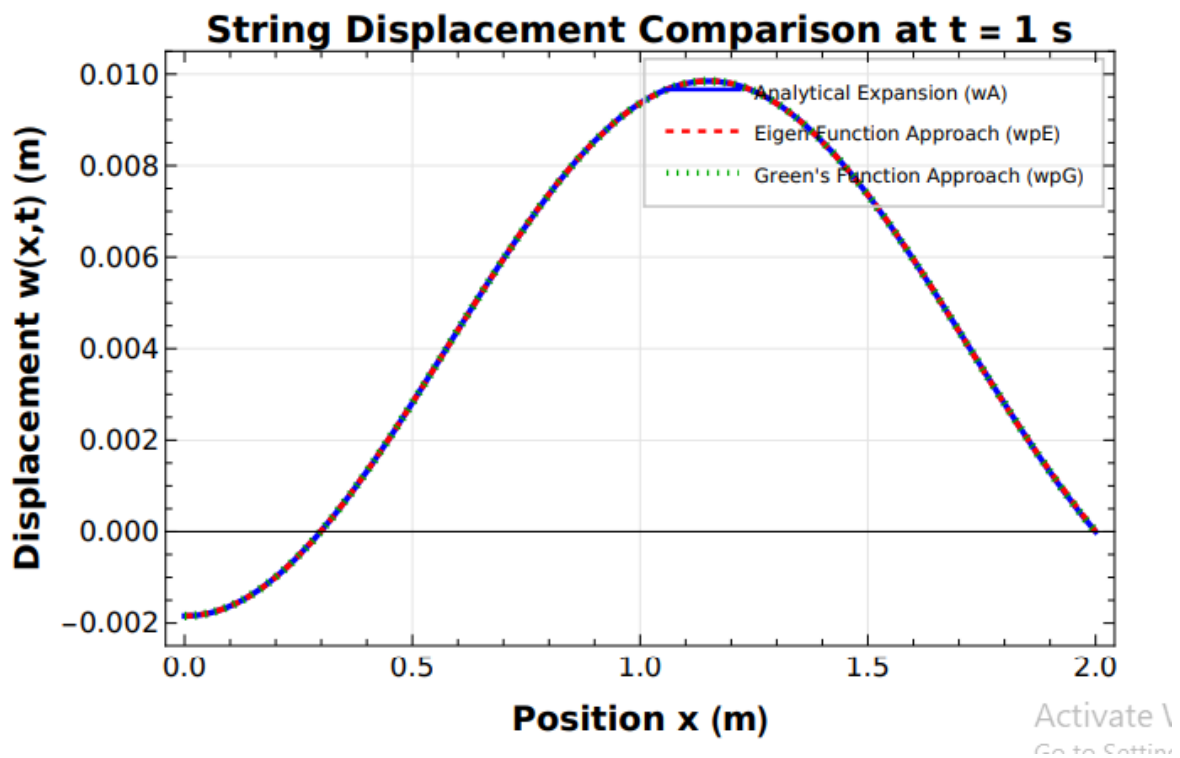
$$\frac{\sec\left(\frac{l\Omega}{c}\right) \cos\left(\frac{\Omega z}{c}\right) \sin\left(\frac{\Omega(l-x)}{c}\right)}{c\Omega}$$

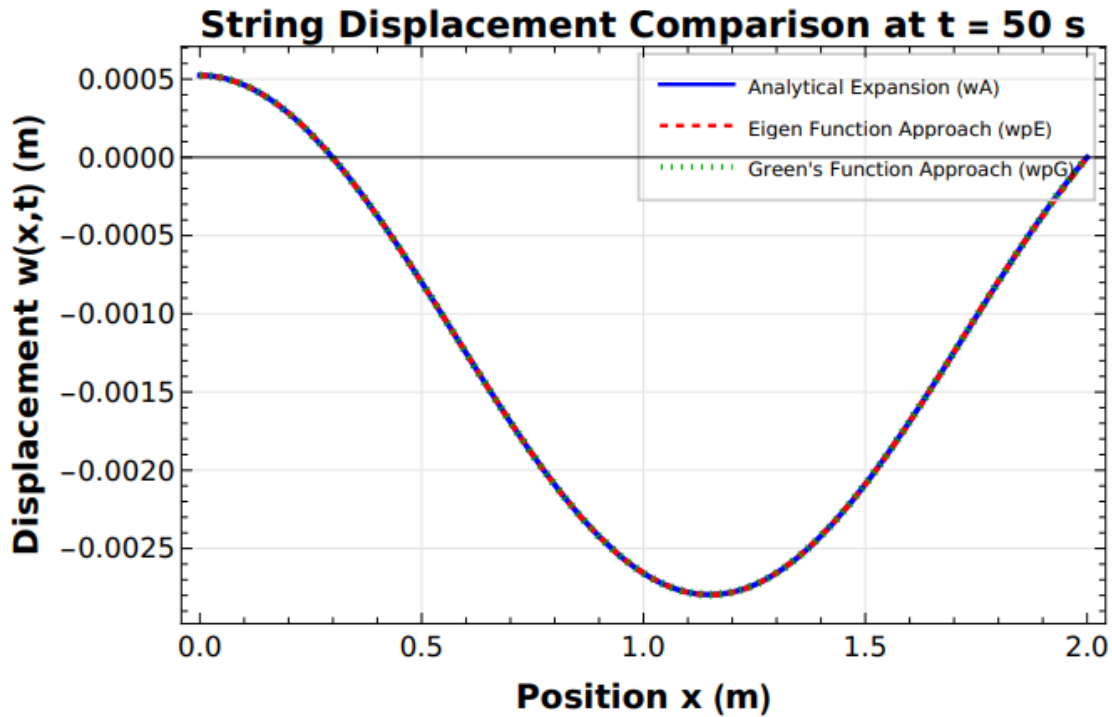
$$\text{Simplified GL: } \frac{\cos\left[\frac{x\Omega}{c}\right] \sec\left[\frac{l\Omega}{c}\right] \sin\left[\frac{(l-z)\Omega}{c}\right]}{c\Omega}$$

$$\text{Simplified GR: } \frac{\cos\left[\frac{z\Omega}{c}\right] \sec\left[\frac{l\Omega}{c}\right] \sin\left[\frac{(l-x)\Omega}{c}\right]}{c\Omega}$$

$$\text{Out[]} = -\frac{2 Q_0 \cos[t \Omega] \sec\left[\frac{l\Omega}{c}\right] \sin\left[\frac{(l-x)\Omega}{2c}\right] \sin\left[\frac{(l+x)\Omega}{2c}\right]}{a l \rho \Omega^2}$$

The Followings the Results of the Displacements in three set of the variable time states:





Code Block for all Logics:

```
Clear["Global`*"];
Q0 = 60; \[Rho] = 7860; a = (Pi*0.25*0.005^2); l = 2; T = 1000;

(* Characteristic constants *)
c = Sqrt[T/(\[Rho] * a)];
\[CapitalOmega] = 220; (* Forcing Frequency rad/s *)
num = 10; (* Number of terms for expansion *)

(* --- 0. Calculating the Steady state solution wA(x,t) --- *)
wA[x_,t_] = -(Q0/(\[Rho]*a*l*\[CapitalOmega]^2))*((Cos[\[CapitalOmega]*x/c]/Cos[\[CapitalOmega]*l/c])-1)*Cos[\[CapitalOmega]*t];
wA[x,t]

(* --- 1. Eigenfunction Expansion Method --- *)
Print[Style["Eigenfunction expansion method", 12, Black, Bold]];

W[k_] := Cos[(2*k-1) * Pi * x / (2* l)];
\[Omega][k_] := (2*k-1) * Pi * c / (2*l);

(* Modal coefficient \[Alpha] *)
\[Alpha][k_] := Integrate[(-Q0/(\[Rho] * a*l)) * W[k], {x, 0, l}] /
```



```

((-\[CapitalOmega]^2 + \[Omega][k]^2) * Integrate[W[k]^2, {x, 0, l}]);
\[Alpha][k]

(* Summing the expansion *)
xpm = Sum[\[Alpha][k] * W[k], {k, 1, num}];

(* General expression used for comparison *)
wpE[x_, t_] := xpm * Cos[\[CapitalOmega] * t]
wpE[x, t]

(* --- 3. Green's Function Method --- *)
Print[Style["Green's function method", 12, Black, Bold]];

(* Define Piecewise Green's Function forms *)
(* 1. Define the Piecewise Green's Function forms *)
GL[x_, z_, \[CapitalOmega]] := AL * Sin[\[CapitalOmega] * x / c] + BL * Cos[\[CapitalOmega] * x / c];
GR[x_, z_, \[CapitalOmega]] := AR * Sin[\[CapitalOmega] * x / c] + BR * Cos[\[CapitalOmega] * x / c];
(* 2. Define the Boundary and Continuity/Jump Conditions *)
(* eq1: Zero slope at x=0 (Free End) *)
eq1 = (D[GL[x, z, \[CapitalOmega]], x] /. x -> 0) == 0;
(* eq2: Fixed at x=l *) eq2 = GR[l, z, \[CapitalOmega]] == 0;
(* eq3: Continuity at the point of application x=z *)
eq3 = GL[z, z, \[CapitalOmega]] == GR[z, z, \[CapitalOmega]];
(* eq4: Slope Jump Condition at x=z *)
eq4 = (D[GR[x, z, \[CapitalOmega]], x] - D[GL[x, z, \[CapitalOmega]], x] /. x -> z) == -1/c^2;
(* 3. Solve for the coefficients *)
sol = Simplify[Solve[{eq1, eq2, eq3, eq4}, {AL, BL, AR, BR}]];

(* 4. Display the results in Symbolic/Traditional Form *)
Print["The Green's Function for 0 <= x <= z is:"];
GL[x, z, \[CapitalOmega]] /. sol[[1]] // Simplify // TraditionalForm
Print["The Green's Function for z <= x <= l is:"];
GR[x, z, \[CapitalOmega]] /. sol[[1]] // Simplify // TraditionalForm

(* Extract and Display the simplified expressions *)
finalGL = Simplify[GL[x, z, \[CapitalOmega]] /. sol[[1]]];
finalGR = Simplify[GR[x, z, \[CapitalOmega]] /. sol[[1]]];
Print["Simplified GL: ", finalGL];
Print["Simplified GR: ", finalGR];
(* Solve for Displacement using Green's Function Integration *)
xpg = (-Q0 / (\[Rho] * a * l)) * (Integrate[finalGR, {z, 0, x}] +
Integrate[finalGL, {z, x, l}]);

wpG[x_, t_] = xpg * Cos[\[CapitalOmega] * t]; wpG[x, t];

```

Github Link of Mathemaica file: https://github.com/sank-55/Acads_Meche_22-26/blob/main/Sem%208/Vibration%20of%20Structures/Vos_assignment3_22ME10083.pdf